

Date: 09/10

$$\begin{aligned}
 x - 3 &= 9 \\
 x &= 9 + 3 \\
 x &= \textcircled{12}
 \end{aligned}$$

$$\begin{aligned}
 4x &= 16 \\
 x &= \frac{16}{4} \\
 &= 4
 \end{aligned}$$

$$\frac{3x}{5} = 7$$

$$3x = 7 \times 5$$

$$x = \frac{7 \times 5}{3}$$

$$3x + 6 = 15$$

$$3x = 15 - 6$$

$$3x = 9$$

$$x = \frac{9}{3}$$

$$= 3$$

$$+5x - 7 = +2x + 8$$

$$5x = +2x + 8 + 7$$

$$5x - 2x = 8 + 7$$

$$3x = 15$$

$$x = \frac{15}{3} = 5$$

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Sarwajay

$$2m - \frac{3}{5} = m + \frac{4}{5}$$

$$2m - m = \frac{4}{5} + \frac{3}{5}$$

$$m = \frac{7}{5}$$

$$\frac{m}{2} - \frac{m}{2} = -6 - 8$$

$$\frac{m}{2} - \frac{2m}{2} = -14$$

$$-\frac{m}{2} = -14$$

$$m = 14 \times 2$$

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$$\frac{MN}{MQ} = \frac{NP}{PQ}$$

$$\begin{aligned} \frac{5}{2.5} &= \frac{7}{PQ} \\ 5 \times PQ &= 2.5 \times 7 \\ 5 \times PQ &= 17.5 \\ PQ &= \frac{17.5}{5} \\ PQ &= 3.5 \end{aligned}$$

$$\begin{array}{r} \overline{35} \\ 5 \overline{) 175} \\ \underline{- 150} \\ 25 \\ \underline{- 25} \\ 0 \end{array}$$

3. In $\triangle MNP$, NQ is a bisector of $\angle N$.
If $MN = 5$, $PN = 7$, $MQ = 2.5$ then
find QP .

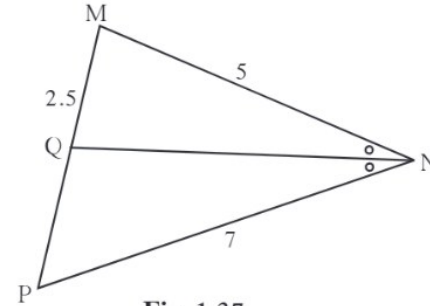


Fig. 1.37

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$$6/x + 6/y = 5 ; 4/x - 3/y = 1$$

$$\frac{6}{x} + \frac{6}{y} = 5$$

$$\frac{4}{x} - \frac{3}{y} = 1$$

$$\text{let } \frac{1}{x} = a$$

$$\frac{1}{y} = b$$

$$6a + 6b = 5$$

$$4a - 3b = 1$$

$$D = \begin{vmatrix} +6 & +6 \\ +4 & -3 \end{vmatrix} = \begin{vmatrix} -12 & -24 \\ -36 & \end{vmatrix}$$

$$D_x = \begin{vmatrix} +6 & +5 \\ -2 & +1 \end{vmatrix} = \begin{vmatrix} 6 & -(-10) \\ 6+10 & \end{vmatrix} = \begin{vmatrix} 16 & \end{vmatrix}$$

$$D_y = \begin{vmatrix} +6 & +5 \\ +4 & +1 \end{vmatrix} = \begin{vmatrix} 6 & -20 \\ -16 & \end{vmatrix}$$

$$a = \frac{D_x}{D}$$

$$= \frac{+16}{-36 - 18}$$

$$= -\frac{4}{9}$$

$$b = \frac{D_y}{D}$$

$$= \frac{+16}{+36 - 9}$$

$$= \frac{4}{9}$$

$$\frac{1}{x} = a, \frac{1}{y} = b$$

$$x = -\frac{9}{4}$$

$$y = \frac{9}{4}$$

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10. एक आयत की लंबाई उसकी चौड़ाई से 3 मीटर अधिक है। यदि आयत का क्षेत्रफल 28 वर्ग मीटर है, तो आयत की लंबाई और चौड़ाई ज्ञात कीजिए।

$$\text{breadth} = x$$

$$\text{length} = x + 3$$

$$\text{Area} = l \times b$$

$$28 = x(x + 3)$$

$$28 = x^2 + 3x$$

$$x^2 + 3x - 28 = 0$$

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Q3. The sum of the digits of a two-digit number is 9. The number obtained by reversing the digits is 27 more than the original number. Find the original number.

$$\begin{array}{l} \text{tens} = x \\ \text{ones} = y \end{array} \Rightarrow \text{number} = 10x + y$$

$$x + y = 9$$

$$10y + x = 27 + (10x + y)$$

$$10y + x = 10x + y + 27$$

$$10y - y + x - 10x = 27$$

$$9y - 9x = 27$$

$$y - x = 3$$